

Bond practice speedwork

Rules: You will have *exactly 5 minutes* to answer each question. After the time is up, you will work with a partner to compare solutions.

Problem.

A \$10,000 par value 10-year bond with 8% annual coupons is bought at a premium to yield an annual effective rate of 6%.

Calculate the interest portion of the 7th coupon.

- A. 632
- B. 642
- C. 651
- D. 660
- E. 667

Problem.

A \$1,000 par value bond bearing 4% annual coupons is purchased at \$887.26 to yield an annual effective rate of 5%.

What is the amount of discount in the second year?

- A. \$4.36
- B. \$4.58
- C. \$4.80
- D. \$5.02
- E. \$5.22

Problem.

A zero-coupon bond with a face value of \$1,000 sells for a price of \$600 and matures in n years, where n is a whole number.

A second bond has the same price, same time until maturity, and same annual effective yield. It is a par-valued bond and pays annual coupons at a rate equal to 50% of the annual effective yield rate.

Calculate the face value of the second bond.

- A. 666.67
- B. 750.00
- C. 774.60
- D. 800.00
- E. 826.40

Problem.

A \$1,000 par value 10-year 8% bond has semiannual coupons. The redemption value equals the par value. The bond is purchased at a premium to yield 6% convertible semiannually.

What is the amount for amortization of the premium in the tenth coupon?

- A. Less than \$7.20
- B. At least \$7.20, but less than \$7.25
- C. At least \$7.25, but less than \$7.30
- D. At least \$7.30, but less than \$7.35
- E. \$7.35 or more

Problem.

Abigail purchases a 5-year \$10,000 bond that pays 3.5% annual coupons and matures at par. She pays \$6,000 for the bond and also deposits the coupons into an account which earns 7.5% interest.

What is her total yield rate on the \$6,000 investment once the bond matures?

- A. 14.1%
- B. 14.3%
- C. 14.5%
- D. 14.7%
- E. 14.9%

Problem.

A bond with coupons equal to 40 sells for P .

A second bond with the same maturity value and term has coupons equal to 30 and sells for Q .

A third bond with the same maturity value and term has coupons equal to 80.

All prices are based on the same yield rate, and all coupons are paid at the same frequency.

Determine the price of the third bond.

A. $4P - 4Q$

B. $4P + 4Q$

C. $4Q - 3P$

D. $5P - 4Q$

E. $5Q - 4P$

Solutions:

1. B
2. B
3. B
4. B
5. E
6. D